

# Benchmarking Regression-Based Cell Type Deconvolution Methods for Bulk RNA-seq Data

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## Abstract

**Motivation:** Determining cell type composition from bulk RNA-seq data is fundamental to computational biology, with direct applications in cancer immunology and disease characterisation. Reference-based deconvolution frames this as a regression problem: given an observed bulk expression vector and a reference signature matrix, recover non-negative cell type proportions. Widely-used tools such as CIBERSORTx and MuSiC embed specific regression engines as black boxes, yet it remains unclear whether the regression algorithm materially affects accuracy or whether upstream factors such as signature matrix quality and gene selection dominate performance.

**Results:** We benchmarked three algorithms—Non-Negative Least Squares (NNLS), Support Vector Regression (SVR), and Robust Regression via Huber loss—on synthetic pseudo-bulk mixtures from 8 immune cell types using GEO dataset GSE107011. All three methods achieve strong accuracy (Pearson  $r > 0.94$ ) with differences of  $\Delta r < 0.003$  across methods. NNLS matches or exceeds competing methods on every accuracy metric, exhibits the greatest noise stability, and is approximately 3 000× faster than SVR. Marker gene selection strategy and cell type transcriptomic distinctness are larger determinants of deconvolution quality than the choice of regression algorithm.

**Availability and Implementation:** All code is implemented in Python 3.10 using `scipy`, `numpy`, and `matplotlib` and is available at <https://github.com/vigneshp/pathak-deconv>.

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## 1. Introduction

Bulk RNA-seq measures aggregate gene expression from tissue samples containing thousands of cells of heterogeneous types. The resulting profile is a weighted mixture of constituent cell type expression programmes, where weights correspond to cell type proportions. Accurate recovery of these proportions is essential for characterising the tumour microenvironment, profiling immune responses, and identifying cellular correlates of disease progression [2].

Reference-based deconvolution solves for proportions by fitting a mixture model: given observed bulk expression  $\mathbf{b}$  and reference signature matrix  $S$  (columns = mean expression per cell type), recover non-negative coefficients  $\mathbf{c}$  such that  $S\mathbf{c} \approx \mathbf{b}$ . Tools such as CIBERSORTx [1], MuSiC [2], EPIC [3], and quanTIseq [4] have made this accessible but embed specific regression engines as opaque implementations.

Although benchmarking studies have compared full deconvolution pipelines [3], few systematically isolate the regression core under controlled conditions. It remains unclear whether algorithm choice substantially affects accuracy, or whether performance is primarily governed by upstream factors such as signature matrix quality and gene set.

We benchmark NNLS, SVR, and Robust Regression across four axes: overall accuracy, noise robustness, cell type scalability, and marker gene sensitivity. Using synthetic pseudo-bulk mixtures with known ground-truth proportions we obtain unambiguous measurements and derive practical recommen-

dations for pipeline design.

## 2. Methods

### 2.1 Data and Preprocessing

We used GEO dataset GSE107011, containing expression profiles from 127 sorted human peripheral blood immune cell samples across 30 fine-grained subtypes. After filtering zero-expression genes and applying  $\log_2$ -transformation, we retained the top 5 000 most variable genes. The 30 subtypes were grouped into 8 major cell types: T cells, B cells, Monocytes, NK cells, Dendritic cells, Neutrophils, Basophils, and Progenitor cells. A reference signature matrix was built by averaging profiles within each major cell type.

### 2.2 Pseudo-Bulk Mixture Generation

Fifty pseudo-bulk mixtures were generated with known ground-truth proportions drawn from a symmetric Dirichlet distribution over 8 cell types. Individual profiles were sampled at random and linearly combined. Gaussian noise scaled to 10% of the per-gene expression standard deviation was added to simulate biological variability; the scale was varied from 0%–100% in robustness experiments.

### 2.3 Marker Gene Selection

Three strategies were evaluated: (i) *Top Variance*—genes ranked by variance across cell types; (ii) *Top Differential Expression*—genes ranked by the difference between highest and second-highest expression across cell types; (iii) *Top Fold Change*—genes ranked by the ratio of maximum to mean ex-

**Table 1.** Overall accuracy across 50 pseudo-bulk mixtures (8 cell types, 10% noise).

| Method       | Pearson $r$ | Spearman $\rho$ | RMSE   | MAE    |
|--------------|-------------|-----------------|--------|--------|
| NNLS         | 0.9436      | 0.9086          | 0.0367 | 0.0241 |
| SVR          | 0.9419      | 0.9062          | 0.0371 | 0.0248 |
| Robust Regr. | 0.9413      | 0.9059          | 0.0374 | 0.0246 |

pression. Gene count was varied from 100 to 5 000.

## 2.4 Deconvolution Methods

**NNLS** minimises  $\|Sc - b\|_2$  subject to  $c \geq 0$ , solved via the active-set algorithm. Coefficients are normalised to sum to one; no hyperparameters are required.

**SVR** uses nu-SVR with a linear kernel, fitting  $S$  as feature matrix to target  $b$ . Cell type coefficients are recovered from the weight vector; negative values are clipped before normalisation.

**Robust Regression** fits Huber-loss regression of  $S$  onto  $b$  without intercept, down-weighting genes with large residuals. This is mathematically analogous to the weighted least squares core of MuSiC.

## 2.5 Evaluation

We computed Pearson  $r$ , Spearman  $\rho$ , RMSE, and MAE from predicted and true proportions flattened across all samples and cell types, plus per-cell-type Pearson  $r$  for each of the 8 types. Wall-clock runtime and peak memory were measured via Python’s .

## 3. Results

### 3.1 Overall Accuracy

All three methods achieve strong accuracy under baseline conditions (8 cell types, 10% noise, 5 000 genes; Table 1). NNLS achieves the highest Pearson  $r$  (0.9436) and lowest RMSE (0.0367), followed by SVR (0.9419) and Robust Regression (0.9413). The maximum difference is  $\Delta r = 0.0023$ , indicating the regression engine is not the primary driver. Figure 1 shows true versus predicted proportions for all three methods.

### 3.2 Per-Cell-Type Performance

Performance varies substantially across cell types (Fig. 2). Basophils, Neutrophils, and Progenitor cells achieve near-perfect accuracy ( $r > 0.99$ ), reflecting highly distinctive transcriptional signatures. T cells and NK cells are intermediate ( $r \approx 0.91$ – $0.92$ ), consistent with shared lymphoid lineage markers. Dendritic cells ( $r \approx 0.87$ ) and Monocytes ( $r \approx 0.85$ ) are the hardest to deconvolve due to myeloid lineage collinearity. This pattern is consistent across all three algorithms.

### 3.3 Noise Robustness

At 100% noise, Pearson  $r$  remains above 0.93 for all methods (Fig. 3). NNLS is most stable, dropping only 0.0092 across the full noise range (0.9437 to 0.9345). Robust Regression drops 0.0065, and SVR degrades the most (0.0103), reaching

**Table 2.** Computational cost (50 samples, 8 cell types).

| Method       | Runtime (s) | Peak Memory (MB) |
|--------------|-------------|------------------|
| NNLS         | 0.04        | 1.04             |
| Robust Regr. | 1.47        | 0.73             |
| SVR          | 118.80      | 0.70             |

$r = 0.9308$  at maximum noise. The NNLS non-negativity constraint acts as an implicit regulariser limiting error accumulation.

## 3.4 Cell Type Scalability

A sharp accuracy drop occurs at 4 cell types ( $r \approx 0.885$ ; Fig. 4), corresponding to the simultaneous inclusion of Monocytes and Dendritic cells, which introduce near-collinearity. Performance recovers at 5–8 types as each additional transcriptomically distinct type improves regression conditioning, converging to  $r \approx 0.94$ .

## 3.5 Marker Gene Selection

At 100 genes, Top Differential Expression achieves  $r = 0.892$  for NNLS, outperforming Top Variance ( $r = 0.853$ ) and Top Fold Change ( $r = 0.762$ ; Fig. 5). Above 2 000 genes all strategies converge to  $r \approx 0.94$ , indicating that selection criterion becomes irrelevant at sufficient gene coverage.

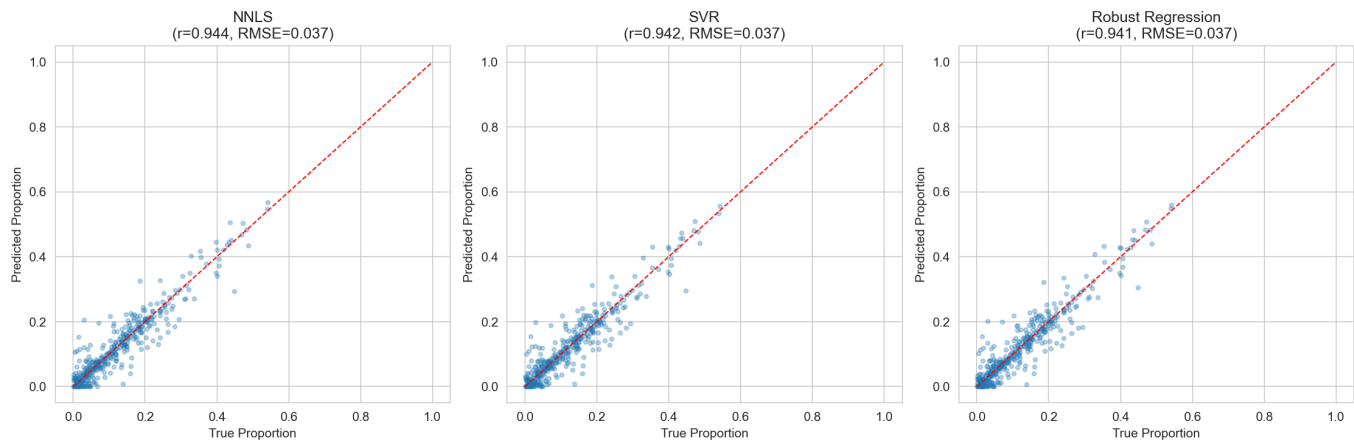
## 3.6 Computational Efficiency

NNLS (0.04 s for 50 runs) is  $37\times$  faster than Robust Regression (1.47 s) and  $2970\times$  faster than SVR (118.8 s; Table 2). Peak memory is comparable across methods (0.70–1.04 MB). SVR’s runtime scales with support vector count, making it impractical for high-throughput studies.

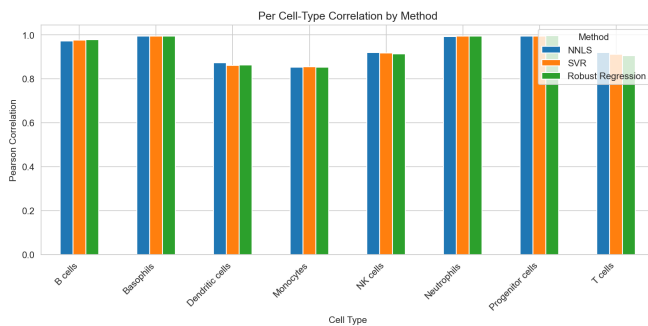
## 4. Discussion

Three algorithms with fundamentally different mathematical formulations produce nearly identical deconvolution accuracy ( $\Delta r < 0.003$ ), strongly suggesting the regression engine is not a meaningful axis of variation at sufficient gene coverage. We recommend NNLS for production pipelines: it matches or exceeds competitors on accuracy, is most noise-stable, and is nearly three orders of magnitude faster than SVR. Its simplicity—single convex optimisation, no hyperparameters—also makes it easy to audit and reproduce.

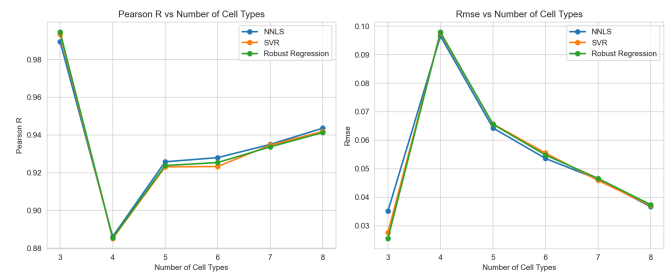
The primary accuracy determinant is not the regression algorithm but the reference signature matrix structure. The Monocyte/Dendritic cell accuracy floor ( $r \approx 0.85$ – $0.87$ ) is consistent across all methods and reflects biological reality: myeloid cells share transcriptional programmes creating irreducible collinearity. Future work should explore condition-number-aware regularisation, hierarchical deconvolution strategies (lineage separation before sub-deconvolution), or lineage-specific marker panels. Marker gene selection is the second most important factor: Top Differential Expression consistently outperforms alternatives when gene budgets are constrained.



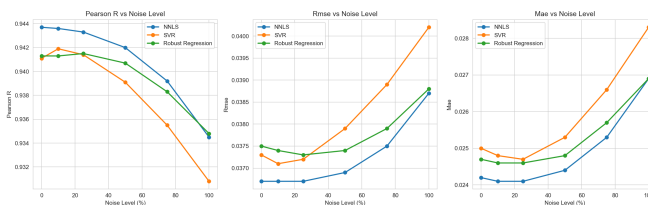
**Figure 1.** True vs. predicted cell type proportions for NNLS (*left*), SVR (*centre*), and Robust Regression (*right*). The dashed line indicates perfect prediction.



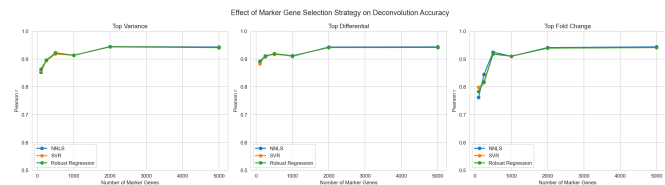
**Figure 2.** Per-cell-type Pearson  $r$  for each method. Myeloid-lineage cells show lowest accuracy due to transcriptomic overlap.



**Figure 4.** Accuracy and RMSE vs. number of deconvolved cell types. The dip at 4 cell types reflects signature matrix collinearity.



**Figure 3.** Pearson  $r$ , RMSE, and MAE vs. noise level (0–100%). NNLS maintains the most stable performance.



**Figure 5.** Pearson  $r$  vs. number of marker genes for three selection strategies (NNLS). Top Differential Expression dominates at low gene counts.

**Limitations.** Synthetic pseudo-bulk mixtures assume linear mixing; real tissue may include cell–cell interactions and spatial heterogeneity not captured here. We tested only 8 cell types; real atlases with 20 or more may amplify collinearity effects. All experiments used a single reference dataset, leaving cross-platform generalisation to future work. Deep learning deconvolution approaches such as Scaden [?] may handle collinear cell types more gracefully and represent a promising direction for future comparison.

## 5. Conclusion

We benchmarked NNLS, SVR, and Robust Regression for reference-based bulk RNA-seq deconvolution across accu-

racy, noise robustness, scalability, and marker gene sensitivity. NNLS achieves the best overall trade-off: highest accuracy ( $r = 0.9436$ ), greatest noise stability, and  $\approx 3\,000\times$  faster execution than SVR. Marker gene selection strategy and cell type transcriptomic distinctness are the dominant determinants of quality. These findings provide actionable guidance: invest effort in signature matrix quality and gene selection rather than in complex regression formulations.

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